

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Replaced ~20 distinct recruitment workflows I'd been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform. Stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.
- Designed and shipped a self-service analytics agent over 67 BigQuery tables — Context Researcher → SQL Drafter → Validator → Devil's Advocate flow with 8 deterministic SQL rules and 23 golden-query test cases. Approved by Sr. Director and product leadership; currently the only AI skill in active use by business teams on Data Ventures. Eliminated the analyst as the bottleneck for routine asks.
- Built an autonomous Jira ticket-resolution agent (6-gate triage, FAISS+BM25 retrieval over 11,000 historical SQL queries, self-healing execution). Closed 400+ data requests across 8 categories over 15 months; cut turnaround from 30–60 minutes per request (and up to a full day for complex ones) to under 10 minutes each.
- Built a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; state persists to BigQuery so the monitor has no memory reliance. Designed a v2 architecture as a 3-tier event-driven platform self-configuring via a BigQuery event registry — new event types ship without code deploys.
- Built a hybrid orchestrator that routes business questions through two backends in parallel — a CubeJS semantic-layer skill (co-built with the DS and product team) and a context-engineered agent — compares outputs, reasons about disagreements, and feeds discrepancies back to refine the semantic layer's definitions over time.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Data: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks, MySQL

Viz: Power BI (DAX, XMLA TMSL), Tableau, LookML, Streamlit, Matplotlib, React Recharts

APIs / triggers / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores

Agent stack: MCP (Model Context Protocol) server design, sub-agent orchestration, context engineering, prompt engineering, RAG (FAISS + BM25 + RRF), ChromaDB semantic memory, multi-model routing across providers, headless cron agents, event-driven architectures

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL